Summary in Graph

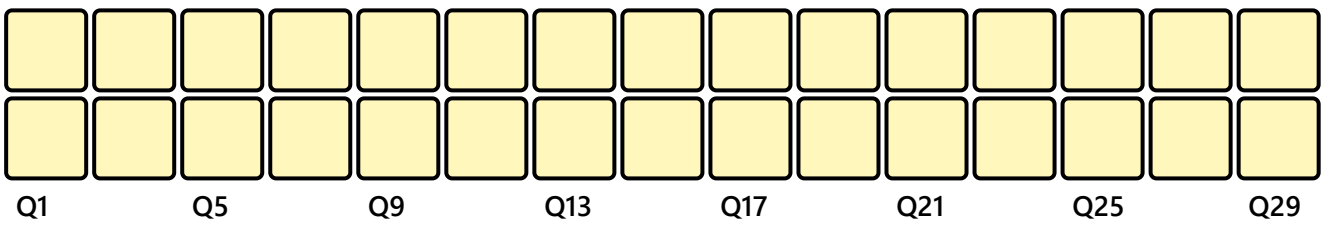
Exam Summary (GATE Overflow CS Test Series 2026 | Operating Systems | Test 2)

Qs. Attempted:	0 (0 + 0)	Correct Marks:	0 (0 + 0)
Correct Attempts:	0 (0 + 0)	Penalty Marks:	0 (0 + 0)
Incorrect Attempts:	0 (0 + 0)	Resultant Marks:	0 (0 + 0)

Total Questions:	30 (10 + 20)
Total Marks:	50 (10 + 40)
Exam Duration:	90 Minutes
Time Taken:	0 Minutes

Individual Question Statistics

Select Metric: Marks Obtained



* Average Stats based on first-attempt stats of Top 10% (5) out of 45 users

EXAM RESPONSE | EXAM STATS | FEEDBACK

Technical

Q #1 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

In a stack page replacement algorithm, the set of pages in a k-frame memory is always a subset of pages in a _____ frame memory (Mark all the appropriate choices)

A. $k-1$
B. $k+1$
C. $k+2$

D. $k(k+1)$

Your Answer: | Correct Answer: B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #2 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a 8-TB disk that uses 2-KB blocks and the free-list method. The number of block addresses that can be stored in one block will be _____

Your Answer: | Correct Answer: 512 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #3 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

Select the correct order of basic page replacement-

I. Location of desired frame is found from disk.
II. Page replacement is used, in case there is no free frame.
III. The process in continued by restarting the instruction caused by trap.
IV. Desired page is bought into newly freed frame and page & frame tables are updated.

A. I, II, III, IV
B. II, I, IV, III
C. I, II, IV, III
D. I, IV, II, III

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #4 | Numerical Type | Award: 1 | Penalty: 0 | Operating System

Consider a file system with 512-byte blocks. Assume an inode of a file holds pointers to 8 direct data blocks, and a pointer to a single indirect block. Further, assume that the single indirect block can hold pointers to 4 other data blocks. The maximum file size in bytes that can be supported by such an inode design will be _____

Your Answer: | Correct Answer: 6144 | Not Attempted | Time taken: 00min 02sec | Discuss

Q #5 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is correct, regarding thrashing? (Mark all the appropriate choices)

A. Increasing the degree of multiprogramming can help in reducing thrashing
B. Priority page replacement can limit the effects of thrashing
C. Thrashing can be avoided by allocating maximum number of page frames to the process needing the maximum amount of memory
D. Thrashing decreases the CPU utilization

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #6 | Multiple Choice Type | Award: 1 | Penalty: 0.33 | Operating System

For the following reference-modify ordered pair, match the following columns-

I.	(Reference bit = 0, Modify bit = 0)	P.	Must write out before replacement
II.	(Reference bit = 0, Modify bit = 1)	Q.	Probably will be used again soon and need to write out before replacement
III.	(Reference bit = 1, Modify bit = 0)	R.	Best page to replace
IV.	(Reference bit = 1, Modify bit = 1)	S.	Probably will be used again soon

A. I - R, II - P, III - S, IV - Q
B. I - R, II - P, III - Q, IV - S
C. I - Q, II - P, III - S, IV - R
D. I - Q, II - P, III - R, IV - S

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #7 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following page replacement algorithms exhibit Belady's anomaly? (Mark all the appropriate choices)

A. Optimal page replacement
B. First in First out (FIFO)
C. Least recently used (LRU)
D. Random page replacement

Your Answer: | Correct Answer: B;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #8 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. Paging avoids external fragmentation
B. Paging avoids internal fragmentation
C. Segmentation avoids external fragmentation
D. Segmentation avoids internal fragmentation

Your Answer: | Correct Answer: A;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #9 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. Dynamic loading is useful in systems with less main memory
B. Dynamic linking reduces the symbol resolution time
C. Dynamic linking does not require special support from the operating system.
D. One of the advantages of static linking is that an application can get benefit from updated libraries without needing a recompilation

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #10 | Multiple Select Type | Award: 1 | Penalty: 0 | Operating System

Which of the following is/are correct? (Mark all the appropriate choices)

A. Both the first-fit and best-fit strategies for memory allocation suffer from external fragmentation.
B. Best-fit strategy for memory allocation avoids internal fragmentation.
C. Best fit is clearly better than first fit in terms of storage utilization.
D. Internal fragmentation exists when there is enough total memory space to satisfy a request but the available spaces are not contiguous.

Your Answer: | Correct Answer: A | Not Attempted | Time taken: 00min 00sec | Discuss

Q #11 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a computer system with enough main memory to hold 4 programs. Assume that these programs wait for I/O 50% of the time. If on average, the fraction of CPU time wasted is a and the degree of multiprogramming necessary to achieve at least 95% CPU utilization is b, the values of $64a + b =$ _____

Your Answer: | Correct Answer: 9 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #12 | Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

Consider a swapping system in which main memory contains the following hole sizes in order in memory : $12K, 14K, 20K, 18K, 7K, 9K, 12K,$ and $15K$. For successive segment requests of $8K, 9K$ and $12K$ holes taken respectively will be (Mark all the appropriate choices)

A. $20K, 18K,$ and $15K$ for Worst-Fit
B. $12K, 14K,$ and $20K$ for First-Fit
C. $12K, 14K,$ and $20K$ for Next-Fit
D. $9K, 12K,$ and $12K$ for Best-Fit

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #13 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A 2048 Kbyte memory is managed using variable partitions but no compaction. It currently has two partitions of sizes 840 Kbyte and 518 Kbyte respectively. The smallest allocation request in Kbyte that could be denied is for _____

Your Answer: | Correct Answer: 231 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #14 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

The address sequence generated by tracing a particular program executing in a pure demand based paging system with 100 records per page with 2 free main memory frames and LRU replacement policy is recorded as follows. The number of page faults will be

$$0100, 0432, 0101, 0612, 0102, 0103, 0104, 0101, 0611, 0102, 0103, 0104, 0101$$

Your Answer: | Correct Answer: 3 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #15 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

Consider a machine with $8\ GB$ physical memory and a 34-bit virtual address space. If the page size s $4\ KB$, the approximate size of the page table assuming 3 auxiliary bits (for protection, page replacement information etc.) on average for each page table entry will be _____

A. $16\ \text{MB}$
B. $8\ \text{MB}$
C. $2\ \text{MB}$
D. $12\ \text{MB}$

Your Answer: | Correct Answer: D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #16 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

In a virtual memory system, size of the virtual address is 64-bit, size of the physical address is 60-bit, page size is $4\ \text{Kbyte}$ and size of each page table entry is 64-bits. The main memory is byte addressable. The maximum number of bits that can be used for storing protection and other information in each page table entry is _____

Your Answer: | Correct Answer: 16 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #17 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a system with byte-addressable memory, 31-bit logical addresses, $4\ \text{kilobyte}$ page size and page table entries of $8\ \text{bytes}$ each. The size of the page table in the system in megabytes is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #18 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A computer system implements huge pages of $1\ \text{megabyte}$ and a 64-bit physical address space. Each page table entry contains a valid bit, a dirty bit, two permission bits, and the translation. If the maximum size of the page table of a process is $48\ \text{megabytes}$, the length of the virtual address supported by the system in bits is _____

Your Answer: | Correct Answer: 43 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #19 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A computer uses 42-bit virtual address, 32-bit physical address, and a three–level paged page table organization. The page table base register stores the base address of the first-level table (T_1), which occupies exactly one page. Each entry of T_1 stores the base address of a page of the second-level table (T_2). Each entry of T_2 stores the base address of a page of the third-level table (T_3). Each entry of T_3 stores a page table entry (PTE). The PTE is $32\ \text{bits}$ in size.
The size of a page in KB in this computer is _____

Your Answer: | Correct Answer: 4 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #20 | Multiple Choice Type | Award: 2 | Penalty: 0.67 | Operating System

If the virtual address space supported is 2^{64} bits, the page size is 4 Kbyte, the size of the physical memory is 2 GB, the size of a PTE is four bytes for standard page table and eight bytes for inverted page table, and the addressing is at the byte level, the sizes of the page tables (assuming single level) required for both standard and inverted page tables respectively will be

A. 2^{54} bytes, 2^{22} bytes
B. 2^{52} bytes, 2^{32} bytes
C. 2^{51} bytes, 2^{22} bytes
D. 2^{50} bytes, 256 bytes

Your Answer: | Correct Answer: C | Not Attempted | Time taken: 00min 00sec | Discuss

Q #21 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A CPU generates 64-bit virtual addresses. The page size is 8 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 256 page table entries and is 8-way set associative. The minimum size of the tag memory for TLB in bytes is _____

Your Answer: | Correct Answer: 1472 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #22 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory (no page faults) and there is no cache. It takes 20 nanoseconds to search the TLB and 300 nanoseconds to access the physical memory. If the TLB hit ratio is 0.9, the effective memory access time (in nanoseconds) is _____

Your Answer: | Correct Answer: 350 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #23 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A typical program has 10% memory instructions. Assume there are 2% data TLB misses, each requiring 100 extra cycles to be handled. Also assume that each instruction requires 1 cycle to execute, data TLB access takes 1 cycle, each memory operation in the physically addressed cache takes 1 cycle, 10% of data accesses are cache misses, and each cache miss takes 45 extra cycles to be handled. How long (in cycles) would it take to execute 100 instructions?

Your Answer: | Correct Answer: 185 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #24 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider the following page reference string:

$$1, 2, 3, 4, 2, 1, 2, 1, 2, 3.$$

Assume demand paging with three frames, and all frames being initially empty. If α is the number of page faults in optimal page replacement, β is the number of page faults in LRU page replacement and γ is number of page faults in FIFO page replacement, the value of $2\alpha + 3\beta + 5\gamma$ is _____

Your Answer: | Correct Answer: 63 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #25 Numerical Type | Award: 2 | Penalty: 0 | Operating System

A demand paging system takes 181 time units to service a page fault and 301 time units to replace a dirty page. Memory access time is 1 time unit. The probability of a page fault is p. In case of a page fault, the probability of page being dirty is also p. It is observed that the average access time is 7 time units. Then the value of $100p$ is _____ (up to two decimal places).

Your Answer: | Correct Answer: 3.15:3.27 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #26 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Consider a computer system with twenty physical page frames. The system is provided with an access sequence $(a_1, a_2, \ldots, a_{2021}, a_1, a_2, \ldots, a_{2021})$, where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is _____

Your Answer: | Correct Answer: 1 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #27 Multiple Select Type | Award: 2 | Penalty: 0 | Operating System

A computer has twenty physical page frames which contain pages numbered 1001 through 1020 entered in order. Now a program accesses the pages numbered $1001, 1002, \ldots, 1100$ in that order, and repeats the access sequence once in the reverse order. Which one of the following page replacement policies experiences the same number of page faults as the optimal page replacement policy for this program? (Mark all the appropriate choices)

A. Least-recently-used
B. First-in-first-out
C. Most-recently-used
D. Last-in-first-out

Your Answer: | Correct Answer: A;B;C;D | Not Attempted | Time taken: 00min 00sec | Discuss

Q #28 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Suppose that a disk drive has 5000 cylinders numbered 0 to 4999. The drive is currently serving a request at cylinder 2150, and the previous request was at cylinder 1805. The queue of pending requests, in FIFO order, is:

$$2069; 1212; 2296; 2800; 544; 1618; 356; 1523;$$

Starting from the current head position, the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for SCAN (Elevator) and Circular SCAN (C-SCAN) disk-scheduling algorithms are a and b respectively, then $2a + 5b =$ ____

Your Answer: | Correct Answer: 64569 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #29 Numerical Type | Award: 2 | Penalty: 0 | Operating System

Suppose that a disk drive has 5000 cylinders, numbered 0 to 4999. The drive is currently serving a request at cylinder 1243, and previously it was at cylinder 125. The queue of pending requests in FIFO order is:

$$86, 1470, 913, 1774, 948, 1509, 1022$$

Starting from the current head position, if the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for First Come First Served (FCFS) and Shortest Seek Time First (SSTF) are α and β respectively, then $2\alpha + 3\beta =$ _____

Your Answer: | Correct Answer: 20303 | Not Attempted | Time taken: 00min 00sec | Discuss

Q #30 | Numerical Type | Award: 2 | Penalty: 0 | Operating System

A computer uses 48-bit virtual address, 32-bit physical address, and a three-level paged page table organization with page size being $16\ \mathrm{KB}$. The processor used in the computer has an $8\ \mathrm{MB},\ 8$ way set associative virtually indexed physically tagged cache. The cache block size is $64\ \mathrm{bytes}$. The minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer is _____

Your Answer: | Correct Answer: 64 | Not Attempted | Time taken: 00min 00sec | Discuss

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